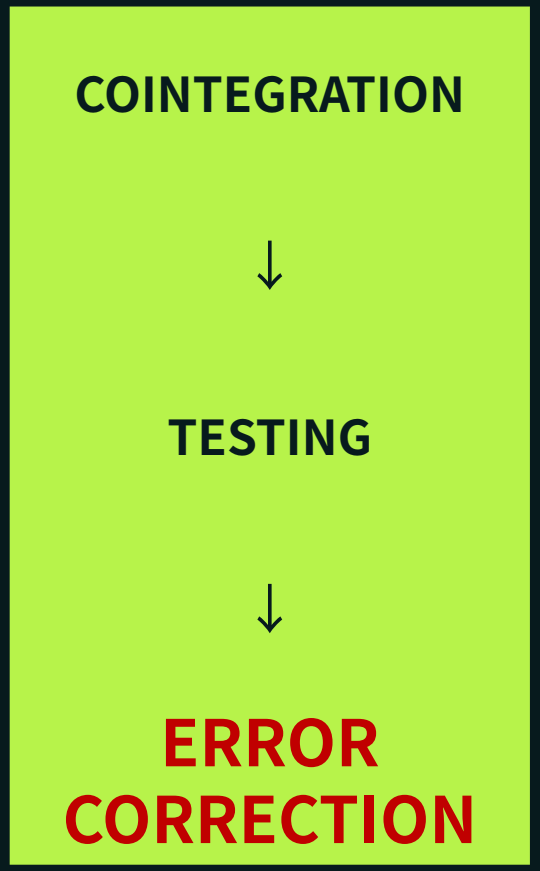
Cointegration & ECM Formula Cheat Sheet

One formula per slide — cointegration · tests · error correction



Advanced Financial Time Series · Week 7

Three parts, formula by formula, in one hour

Each part runs definition → tests and models → pictures, one formula per slide

01	**Cointegration**	**A stationary mix of two wandering series** I(1) · definition · common trend · spurious
02	**Testing for cointegration**	**Prove it with a test, not a chart** EG four-step decisions · critical values · Johansen in practice
03	**Error correction and trading**	**Measure the pull back, then trade it** ECM · VECM · half-life · z-score · breakdown
04	**Synthesis**	**Concept → test → trade** Tests compared · pairs-trading pipeline

Find the anchor with cointegration → prove it with a test → trade it with error correction

01

Cointegration

Two series that each wander, but one combination stays put

A random walk is I(1); its difference is I(0)

Order of integration: how many differences make it stationary — the basic idea behind cointegration

Random walk and its difference

$$y_t = y_{t-1} + \varepsilon_t \;\Rightarrow\; y_t \sim I(1)$$
$$\Delta y_t = y_t - y_{t-1} = \varepsilon_t \sim I(0)$$

How to read it

- I(1): shocks pile up, no mean to return to — price levels
- I(0): moves around a fixed mean — returns
- Differencing makes it stationary but throws away the levels
- Week 6's VAR used differenced series

Cointegration recovers the long-run link in levels that differencing threw away

Cointegrated: a stationary mix of two I(1) series

Definition by Engle and Granger (1987) — Nobel Prize in Economics, 2003

Definition

$$X_t \sim I(1), \quad Y_t \sim I(1), \qquad z_t = Y_t - \beta X_t \sim I(0)$$

Cointegrating vector (1, −β)

$$z_t = \underbrace{\begin{pmatrix} 1 & -\beta \end{pmatrix}}_{\text{cointegrating vector}} \begin{pmatrix} Y_t \\ X_t \end{pmatrix} = Y_t - \beta X_t$$

How to read it

- X and Y each wander, but the spread z is stationary
- It holds for one particular β only
- z is the equilibrium error — the spread we watch
- In trading, β becomes the hedge ratio

The heart of cointegration analysis is finding the β that makes the spread stationary

A stationary mix needs one shared trend

Mixing two I(1) series usually gives I(1) again — unless they share the same stochastic trend

Usually impossible

$$X_t \sim I(1),\ Y_t \sim I(1) \implies aX_t + bY_t \sim I(1) \quad \text{(in general)}$$

Common-trend form

$$\tau_t = \tau_{t-1} + \eta_t \qquad \text{(common stochastic trend)}$$
$$X_t = \tau_t + u_t, \qquad Y_t = \beta\,\tau_t + v_t$$
$$z_t = Y_t - \beta X_t = v_t - \beta\,u_t \ \sim\ I(0)$$

How to read it

- τ: the random walk both series share
- u, v: their own stationary noise
- In Y − βX, the βτ cancels exactly
- Cointegration = fewer trends than series

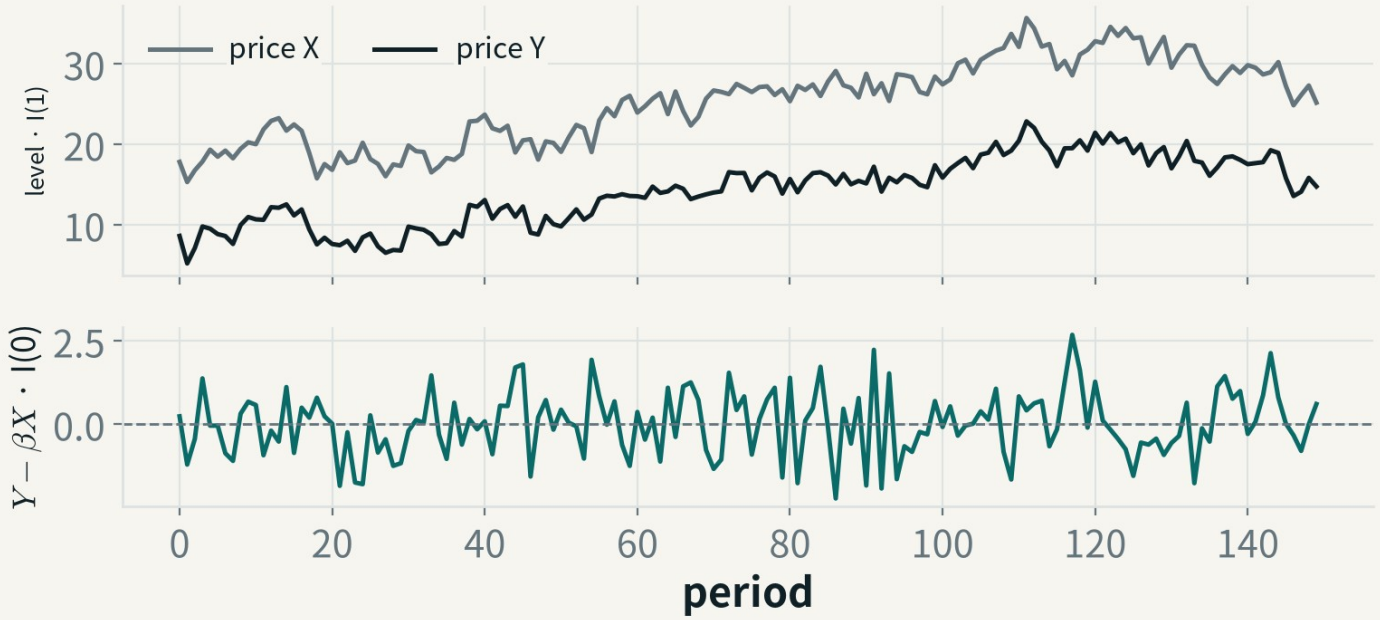
If each has its own trend, no combination can remove them

Prices wander, but the spread returns to 0

A drunk and her dog: each a random walk, but the leash ties their distance (simulation, illustrative data)

THE LEASH

- Top: price levels, each I(1)
- Bottom: the spread, I(0)
- Estimated $\beta \approx 0.79$
- Tied together, but not fixed



A gap that returns to its mean is the long-run equilibrium

Stationary residuals mean the regression is not spurious

The exact exception to Week 2's warning 'do not regress trends on trends'

The residuals decide

$$Y_t = \alpha + \beta X_t + e_t : \quad \begin{cases} e_t \sim I(1) & \Rightarrow \text{ spurious} \\ e_t \sim I(0) & \Rightarrow \text{ cointegrated} \end{cases}$$

How to read it

- Spurious: high R^2, but the residuals wander
- Cointegrated: stationary residuals — proof of a real link
- So testing the residuals for stationarity tells them apart
- That test is Part 2's Engle-Granger

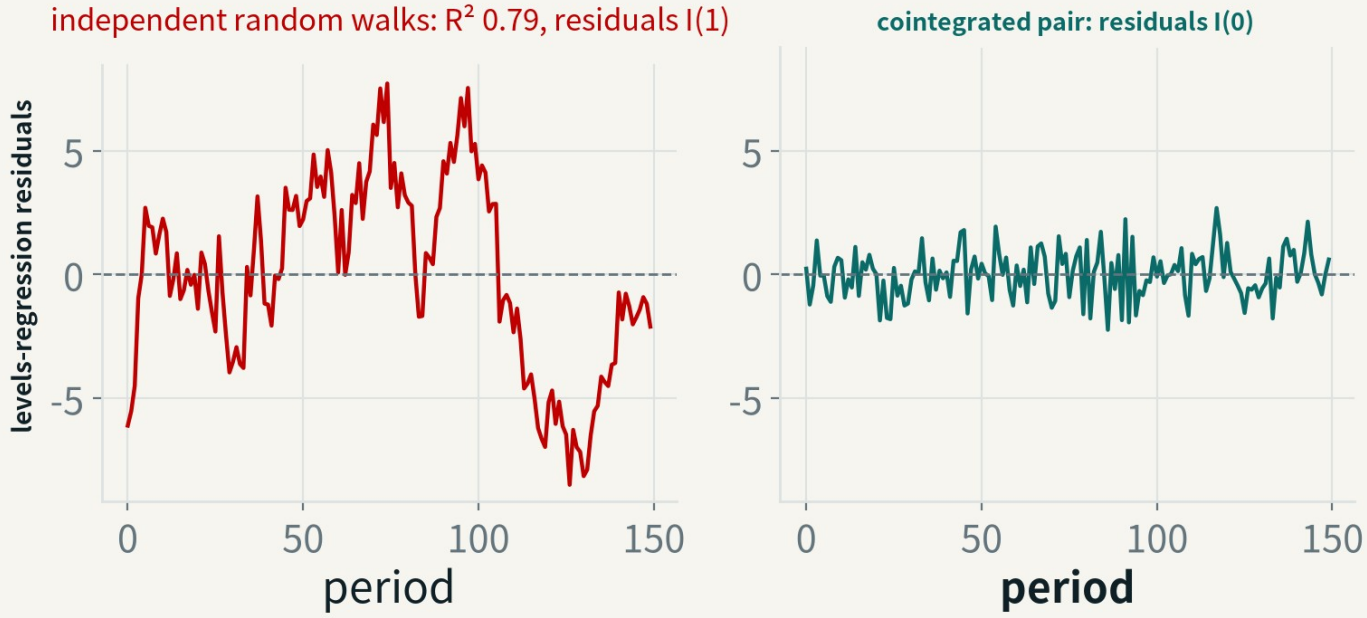
Stationary residuals certify a real relationship

Spurious residuals wander; cointegrated ones return

Same vertical axis, 150 periods of levels-regression residuals (simulation, illustrative data)

SPURIOUS vs GENUINE

- Even two independent random walks get $R^2 = 0.79$
- But the residuals drift far to one side
- Cointegrated residuals stay in a narrow band around 0



Look at the residuals, not R^2

Not correlation — and the spread is a trading signal

Correlation: returns move together short term · cointegration: price levels share a long-run equilibrium

Correlation vs cointegration

$$\rho = \mathrm{Corr}(\Delta X_t,\, \Delta Y_t) \quad \text{vs.} \quad Y_t - \beta X_t \sim I(0)$$

The spread as a trade

$$z_t > \mu_z : \ \text{short } Y,\ \text{long } \beta X$$
$$z_t < \mu_z : \ \text{long } Y,\ \text{short } \beta X$$

How to read it

- High correlation ≠ cointegration
- Candidates: spot–futures, dual listings, same-driver firms
- No economic tie, no cointegration
- A non-pair's 'spread' is a random walk

Market-neutral — buy one, sell the other

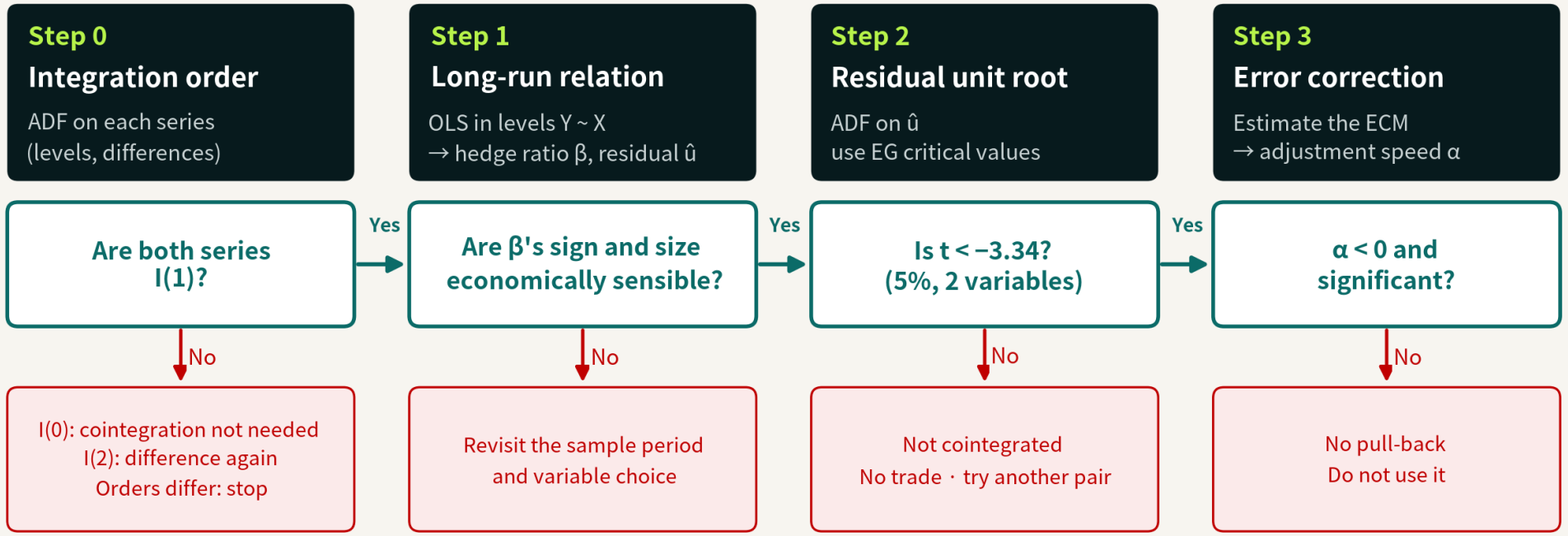
02

Testing for cointegration

Engle-Granger four-step decisions and Johansen in practice

Engle-Granger has four steps; each needs a 'yes' to go on

Decision order for a pair (Y, X) — at any 'no', stop and act as below



Use the spread as a trading signal only after four 'yes'es (Part 3)

Step 0: test for cointegration only if both series are I(1)

ADF on levels and on first differences — null: 'a unit root (non-stationary)'

I(1): levels not rejected, differences rejected

$$X_t \sim I(1) \iff \begin{cases} \text{ADF on } X_t: & \text{not reject} \\ \text{ADF on } \Delta X_t: & \text{reject} \end{cases}$$

How to read it

- Both I(1) → go to step 1
- Levels already rejected (I(0)) → plain regression; no cointegration needed
- Differences not rejected (I(2)) → difference once more, restart at step 0
- Only one is I(1) → no cointegration; stop here

Step 0 asks: is cointegration even worth testing?

Step 1: OLS on levels gives the hedge ratio and the spread

This is estimation, not a test — ignore its t-values and R^2

Levels regression

$$\text{Step 1:} \quad Y_t = \alpha + \beta X_t + u_t \xrightarrow{\text{OLS}} \hat{u}_t = Y_t - \hat{\alpha} - \hat{\beta} X_t$$

How to read it

- β: hedge ratio — for 1 unit of Y, β of X the other way
- û: estimated equilibrium error = the spread
- If β has an odd sign or size, recheck sample and variables
- Run both directions, Y~X and X~Y

Step 1 yields just two things: hedge ratio and residuals

Step 2: stationary residuals mean cointegration; otherwise stop

Null: 'not cointegrated (residual unit root)' — reject it to claim cointegration

ADF on residuals

$$\Delta \hat{u}_t = \gamma\, \hat{u}_{t-1} + \sum_{i=1}^{p} \phi_i\, \Delta \hat{u}_{t-i} + e_t, \qquad H_0 : \gamma = 0$$

Decide by EG critical values

$$t_\gamma = \frac{\hat{\gamma}}{\mathrm{se}(\hat{\gamma})} \;<\; c_{\mathrm{EG}} \;\Rightarrow\; \text{reject } H_0 \;\Rightarrow\; \text{cointegrated}$$

How to read it

- t < −3.34 (5%, 2 variables) → cointegrated, go to step 3
- t ≥ −3.34 → not cointegrated; do not trade the pair
- The ordinary ADF value −2.86 finds false cointegration
- Stricter with 3 variables (−3.74) or 4 (−4.10)

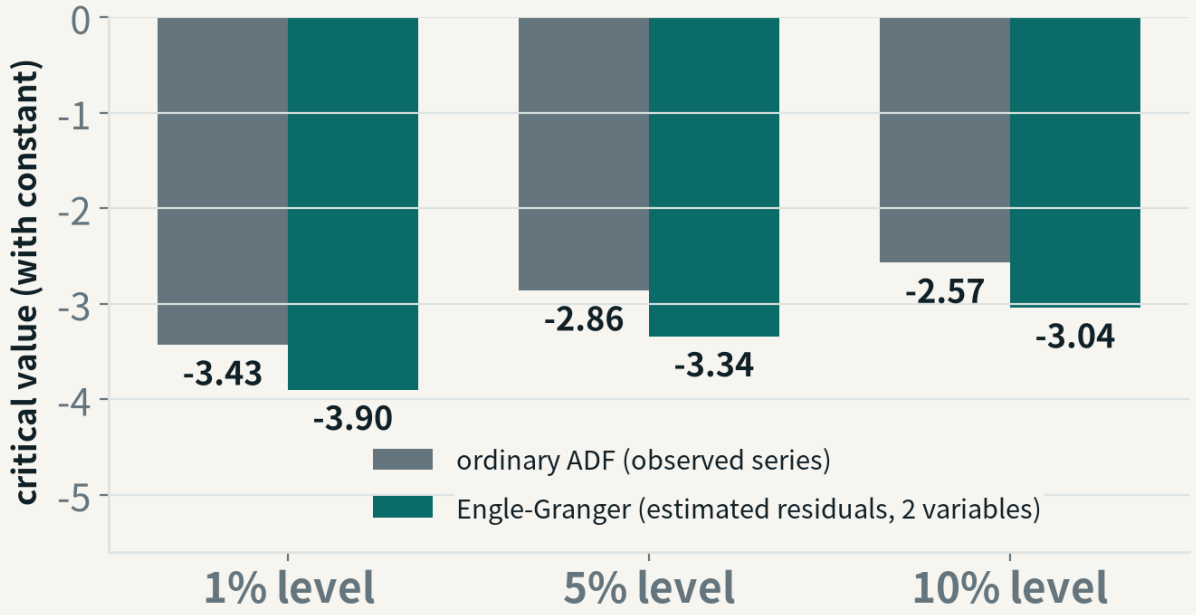
A t between −2.86 and −3.34 is the most common trap

Estimated residuals need stricter critical values

MacKinnon asymptotic critical values, with constant — more negative with more variables

CRITICAL VALUES

- OLS picks β to make the residuals small
- So the residuals look more stationary than they are
- So we raise the bar to correct for it



Never use ordinary ADF critical values in step 2

Step 3: check that a widening gap narrows next period

Error-correction model (ECM) — does last period's spread explain today's change?

ECM and adjustment speed α

$$\Delta Y_t = \alpha\, \hat{u}_{t-1} + \gamma\, \Delta X_{t-1} + \varepsilon_t, \qquad \alpha < 0$$

How to read it

- α < 0 and significant → it pulls back; a confirmed candidate
- α ≈ 0 → cointegrated on paper, but too slow to be useful
- α gives the half-life (Part 3)
- α > 0 means the gap widens — suspect the model

Passes the test + fast pull back = a tradable pair

Three or more series: Johansen — five steps in practice

Leave VECM theory to software; we set two inputs and read two outputs

Step	To do	Decision · next action
0	Check each series is I(1)	Same as EG step 0 — only I(1) goes in
1	Pick lag p with a levels VAR	p minimising AIC · BIC; use p − 1 in Johansen
2	Trace test from r = 0	Reject → next r; otherwise stop
3	Stop r = rank	0: no relation · 1 or more: r spreads
4	Spread from the first vector	Check ADF and half-life, then Part 3

r = k (all variables) means they were stationary already — back to step 0

The trace statistic sums eigenvalues, times T, against critical values

Software gives the eigenvalues $\hat{\lambda}$ — by hand you only add, multiply and compare

Trace statistic (T: observations, k: variables)

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{k} \ln\left(1 - \hat{\lambda}_i\right)$$

What rank r means

$$r = 0: \ \text{no relation} \qquad r = k: \ \text{all } I(0)$$
$$0 < r < k: \ r \text{ cointegrating relations}$$

How to read it

- Null: at most r relations
- 5% critical values (k − r = 1, 2, 3, 4): 3.84 · 15.49 · 29.80 · 47.85
- Statistic > critical value → reject, next r
- The first r not rejected is the rank

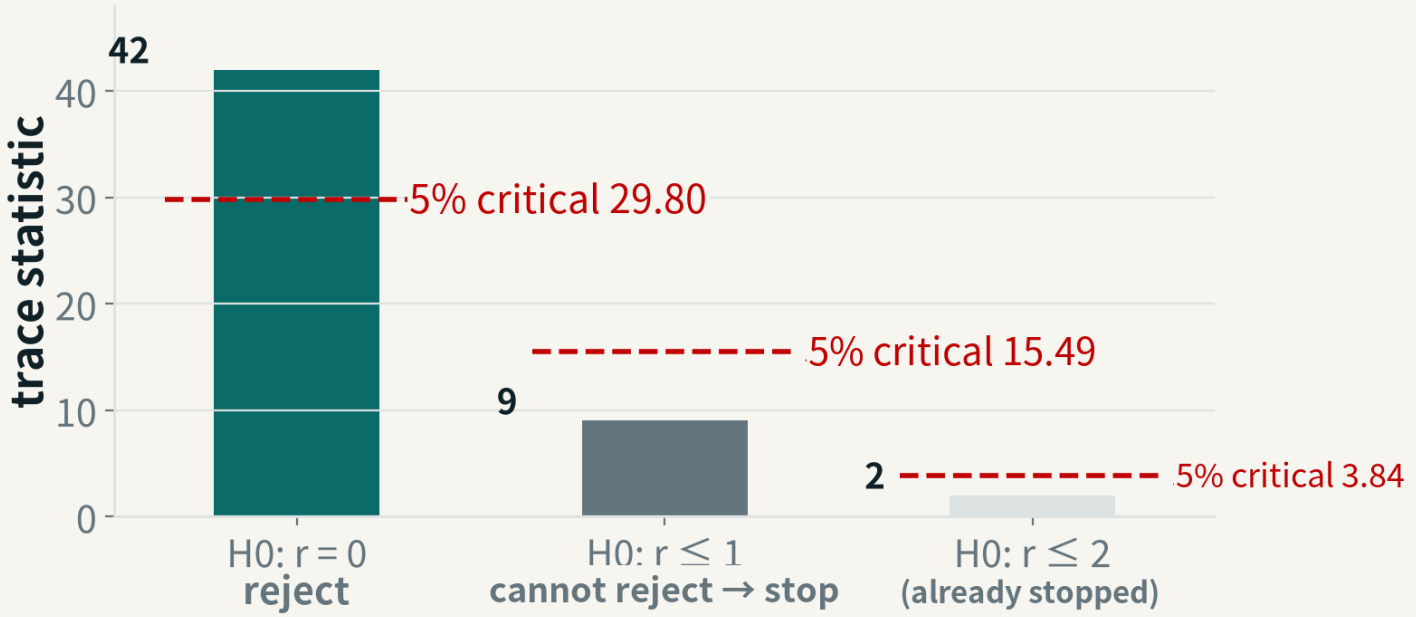
Climb up from r = 0; the first stop is the answer

Only r = 0 is rejected: the rank is 1

Trace statistics for three series (about 42, 9, 2 from the course figure) and standard 5% critical values

READING THE TRACE TEST

- 42 > 29.80 → reject r = 0
- 9 < 15.49 → cannot reject r ≤ 1
- Stop here: rank 1
- One independent long-run equilibrium



For a single pair, the Johansen rank must be 0 or 1

With rank 1, the first cointegrating vector is the spread to trade

Normalise the first element to 1: the rest are hedge ratios per unit of Y1

Spread from the normalised coint. vector

$$z_t = \hat{\beta}^\top \mathbf{Y}_t = Y_{1,t} + \hat{\beta}_2 Y_{2,t} + \cdots + \hat{\beta}_k Y_{k,t}$$

How to read it

- Python: coint_johansen in statsmodels
- Input: det_order 0 (with constant), k_ar_diff = p − 1
- Output lr1: trace statistics; cvt: 90 · 95 · 99% critical values
- Output evec, first column: first coint. vector

Hold assets by the vector coefficients, opposite ways: the spread

A pair: Engle-Granger. A system: Johansen

Both ask the same question — 'is there a stationary combination?' — in different ways

Feature	Engle-Granger	Johansen
Series	2 (a pair)	2 or more
Approach	One equation, 4 steps	System (VECM-based)
Relations found	At most 1	At most k − 1 (the rank)
Symmetric?	No — pick a dependent	Yes

※ Traps: structural breaks, short samples (low power), lag choice — check robustness across lags, orderings and subsamples

A positive result gives β and the spread — the raw material for error correction and trading

03

Error Correction and Pairs Trading

Measure the force that pulls the spread back, then trade it

The ECM explains the next change by last period's gap

Two-variable ECM — a gap above 0 is pulled down, below 0 pulled up

Two-variable error-correction model (ECM)

$$\Delta Y_t = \alpha_Y \underbrace{(Y_{t-1} - \beta X_{t-1})}_{z_{t-1}\ (\text{last gap})} + \gamma_Y \Delta X_{t-1} + \varepsilon_{Y,t}$$

$$\Delta X_t = \alpha_X (Y_{t-1} - \beta X_{t-1}) + \gamma_X \Delta Y_{t-1} + \varepsilon_{X,t}$$

How to read it

- $z(t-1) = Y - \beta X$: last period's equilibrium error
- α: how fast each series reacts to the gap
- γ terms: short-run effects, separate from the long run
- Every term is stationary, so no spurious regression

No error-correction term, no cointegration

A VECM is a differenced VAR plus one correction term

It puts back, as an explicit correction term, the long run that differencing drops

VECM

$$\Delta \mathbf{Y}_t = \alpha \underbrace{\left(\beta^\top \mathbf{Y}_{t-1}\right)}_{z_{t-1}} + \sum_{i=1}^{p-1} \Gamma_i \, \Delta \mathbf{Y}_{t-i} + \boldsymbol{\varepsilon}_t$$

The spread's own adjustment speed

$$\Delta z_t = \Delta Y_t - \beta \, \Delta X_t = \underbrace{(\alpha_Y - \beta \, \alpha_X)}_{\alpha_z < 0} \, z_{t-1} + \cdots$$

How to read it

- Γ terms: short-run momentum and noise (an ordinary VAR)
- α(β′Y) term: all of the long-run behaviour
- $\alpha < 0$ is needed for the gap to close
- The series with larger $|\alpha|$ does most of the correcting

The only difference from Week 6's VAR is this correction term

The half-life turns adjustment speed into days

Ignoring short-run terms, the spread is AR(1) — a share ρ of the gap remains each period

The spread as an AR(1)

$$z_t = (1+\alpha_z)\, z_{t-1} + u_t = \rho\, z_{t-1} + u_t$$
$$-1 < \alpha_z < 0 \iff 0 < \rho < 1$$

Half-life

$$\rho^h = \tfrac{1}{2} \implies h_{1/2} = \frac{\ln 0.5}{\ln(1+\alpha_z)} = \frac{\ln 0.5}{\ln \rho}$$

How to read it

- ρ = 0.90 (α = −0.10) gives a half-life of about 6.6 days
- Sets a sensible holding period
- Screens out spreads that are too slow
- Checks the estimated speed is plausible

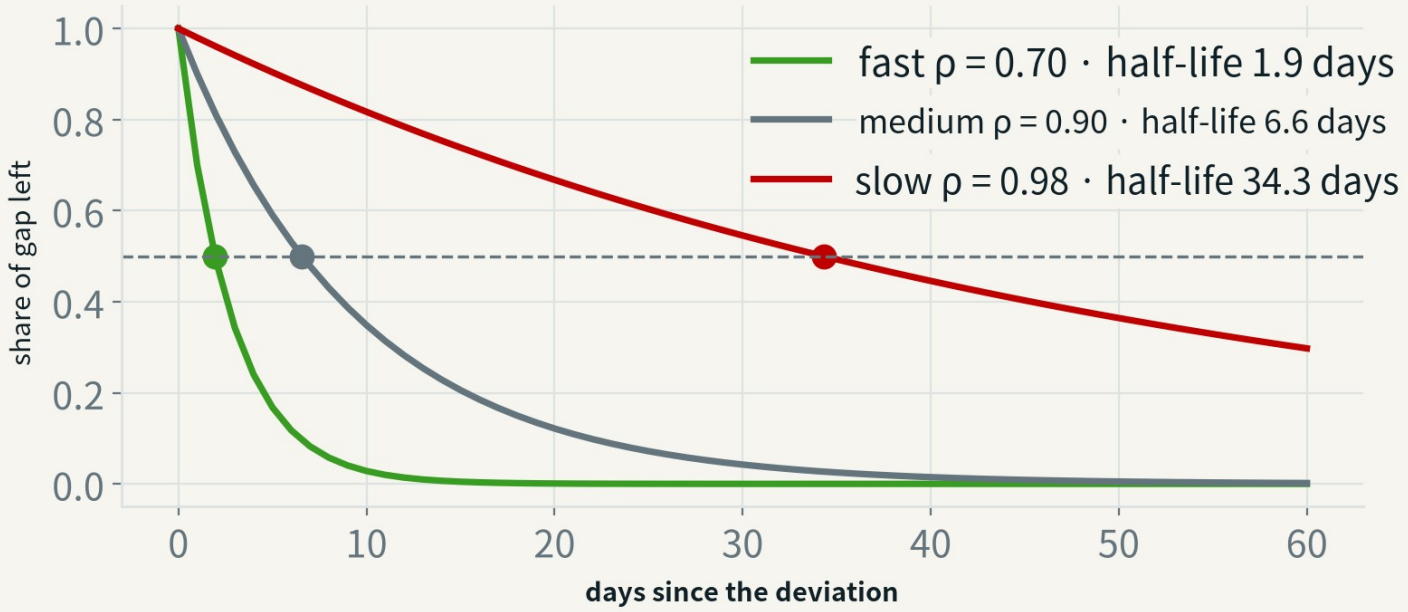
Between 3-day and 60-day half-lives, the 3-day one is more tradable

Half the gap closes in 2 days at ρ 0.70, 34 at 0.98

Share of the gap left after a deviation; dots mark each half-life

FAST vs SLOW

- Fast ρ 0.70: 1.9 days
- Medium ρ 0.90: 6.6 days
- Slow ρ 0.98: 34.3 days
- The slower, the longer capital is tied up



Fast, reliable reversion is what makes a trade worth doing

Cointegration and ECM come together; signal by z-score

Granger representation theorem → entry and exit rules for pairs trading

Granger representation theorem

$$\text{cointegrated} \iff \text{error-correction representation exists}$$

z-score signal

$$Z_t = \frac{z_t - \mu_z}{\sigma_z}, \qquad |Z_t| \geq 2: \text{ enter}, \qquad Z_t \to 0: \text{ exit}$$

How to read it

- Never one without the other — why we use a VECM
- $Z \geq +2$: sell the spread (sell Y, buy βX)
- $Z \leq -2$: buy the spread; exit when Z returns to 0
- Estimate μ and σ from a past window only (no look-ahead)

Bet against the stretch, collect when it snaps back

Enter at ±2σ, exit at 0: P&L builds in steps

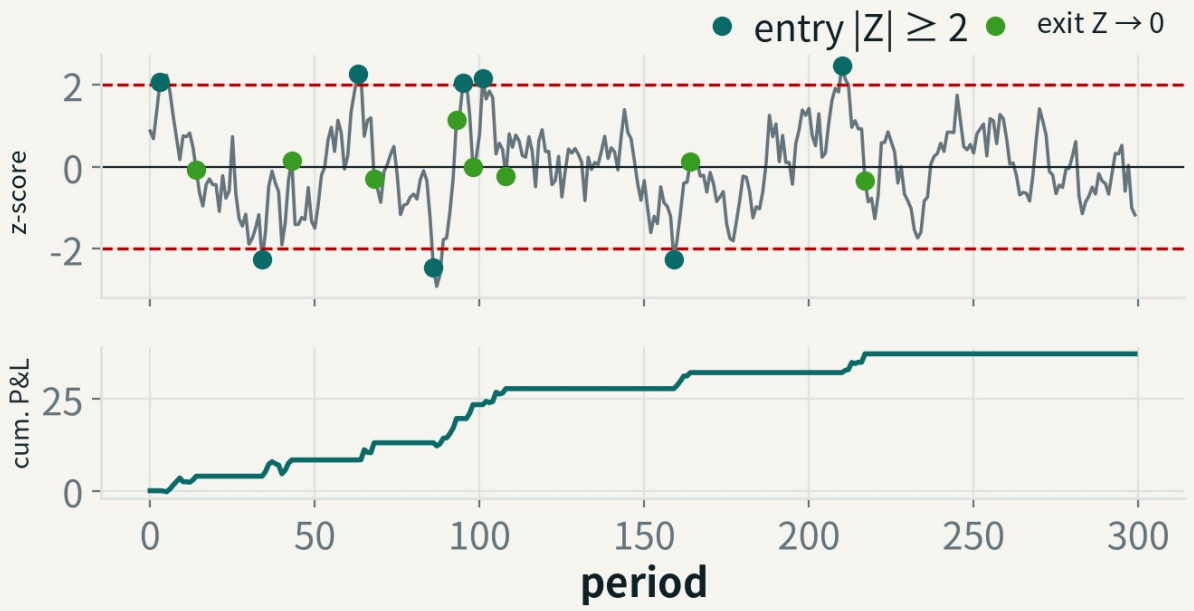
Simulated spread with ρ = 0.9, 300 periods, 8 entries, P&L in spread units (illustrative data)

Position and P&L

$$\text{position} = \pm\,(1 \text{ unit of } Y,\ -\beta \text{ units of } X)$$
$$\text{P\&L}_t = \text{pos}_{t-1}\,\Delta z_t, \qquad \text{size} \propto \sigma_z^{-1}$$

PAIRS TRADING STAGE II

- Market-neutral with the β ratio
- Size by spread volatility



The bet is on the spread, not the market

When the relation breaks, the spread drifts away

Mean-reverting AR(1) at first, a trending random walk after t = 180 (illustrative data)

WHEN IT BREAKS

- Causes: mergers, regulation, new business models
- The old β no longer fits
- Defences: stop-loss, re-testing, many pairs
- Cointegration is an assumption, not a law



Pairs trading without a stop-loss is a bet on a random walk

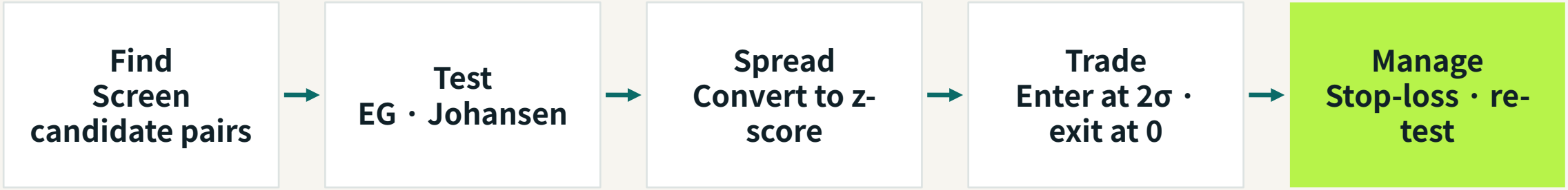
04

Synthesis

Concept → test → trade, and the ever-present risk of breakdown

Three steps: find, prove, trade

Pairs-trading pipeline — engine below: cointegration (anchor), error correction (pull back), volatility (bands and size)



Part	Key idea
Part 1 Cointegration	One common trend → stationary spread = long-run equilibrium
Part 2 Tests	EG residual test · Johansen rank
Part 3 Trading	α · half-life · z-score · breakdown risk

Next, Week 8: state-space models and the Kalman filter

Find the anchor — cointegration

Prove the anchor — EG · Johansen

Trade the pull back — error correction

Next week
State-space models and the Kalman filter